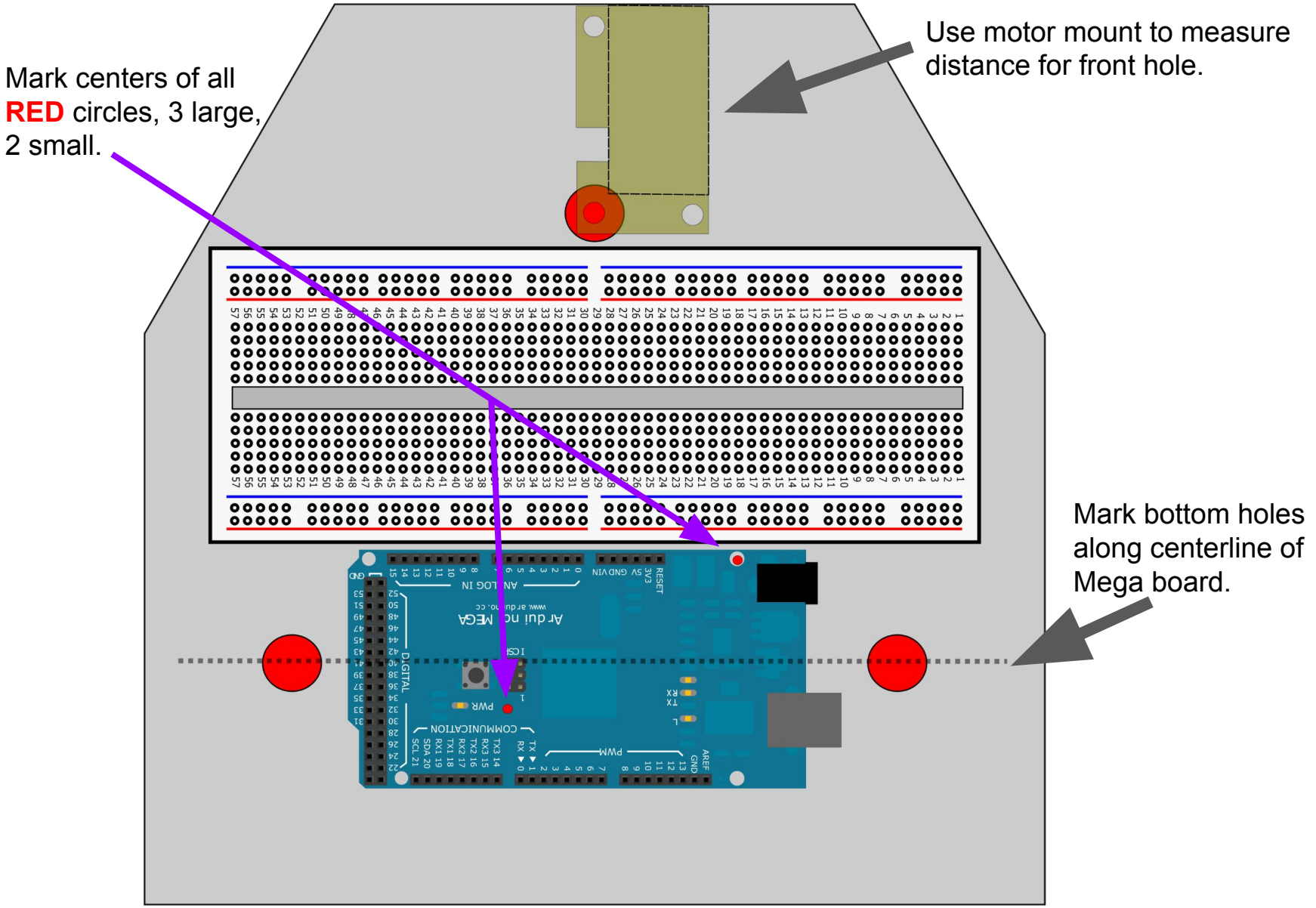
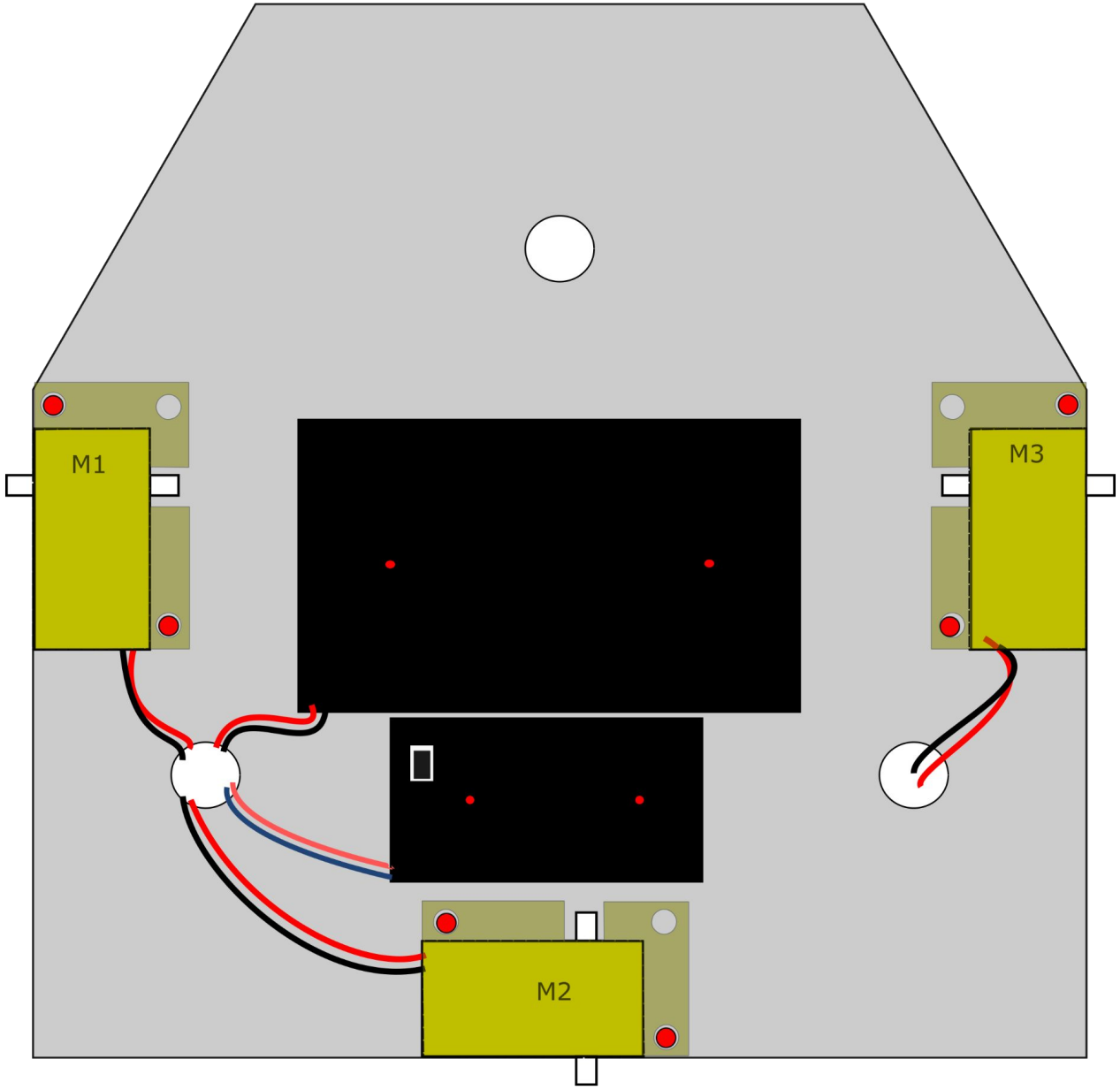
BreadBoardBot

Layout and wiring, rev. 2025b
6/11/2025

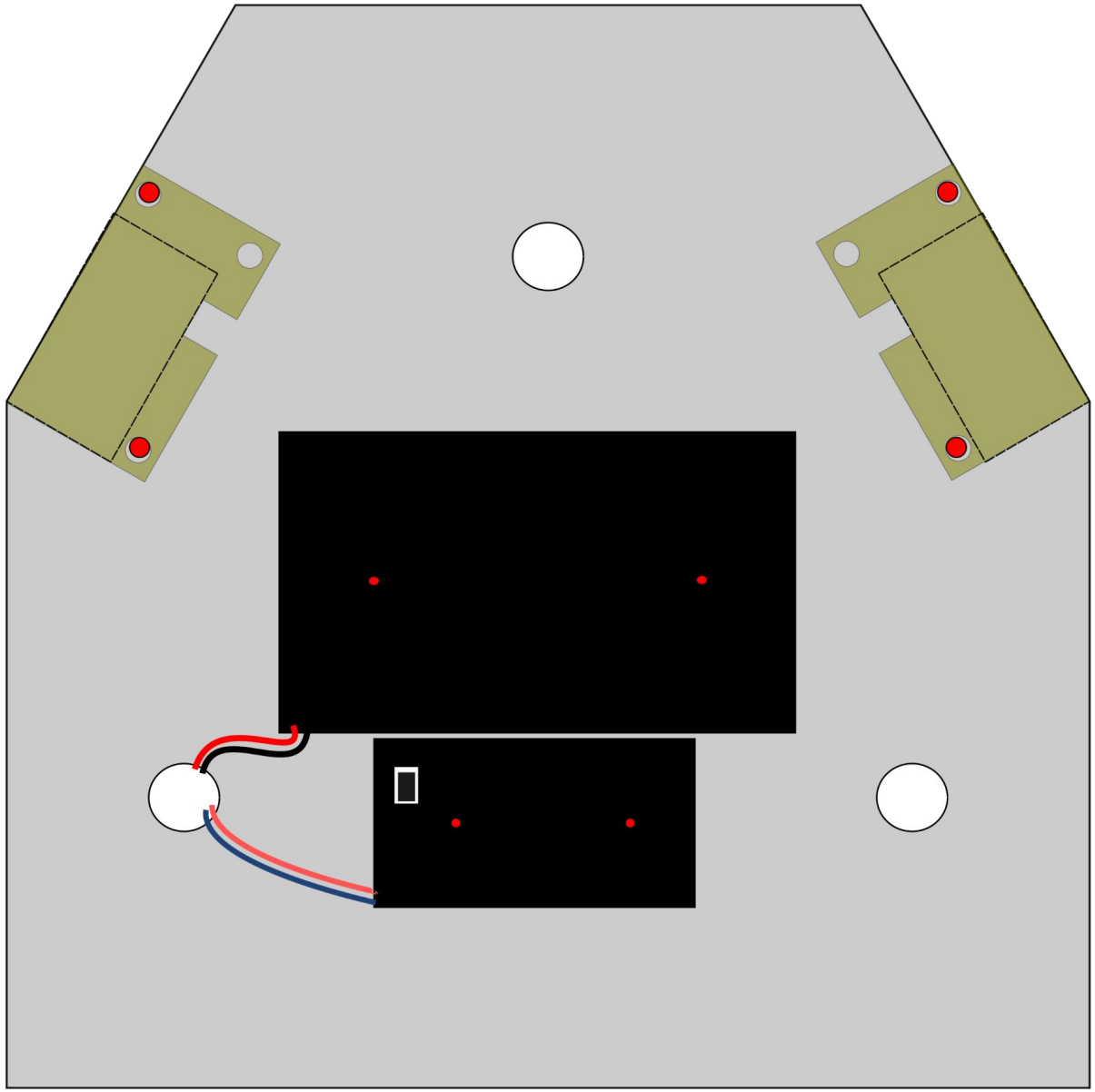
B3 Top Layout: 3 large holes, 2 medium holes



B3 Bottom Layout 1: 10 small holes



B3 Bottom Layout 2: 4 more small holes



Drill holes, then…

- **Always wear safety glasses when using the drills.** This is non-negotiable.
- Use medium sized bit for all motor mount holes (10 of those) and, HIGHLY SUGGESTED, to drill pilot holes for the ½” big holes (3 of those).
- Drill six small holes, 2 marked on front for the Mega, and 4 on the back for both battery holders.
- Drill the three big holes, which is much easier with pilot holes.

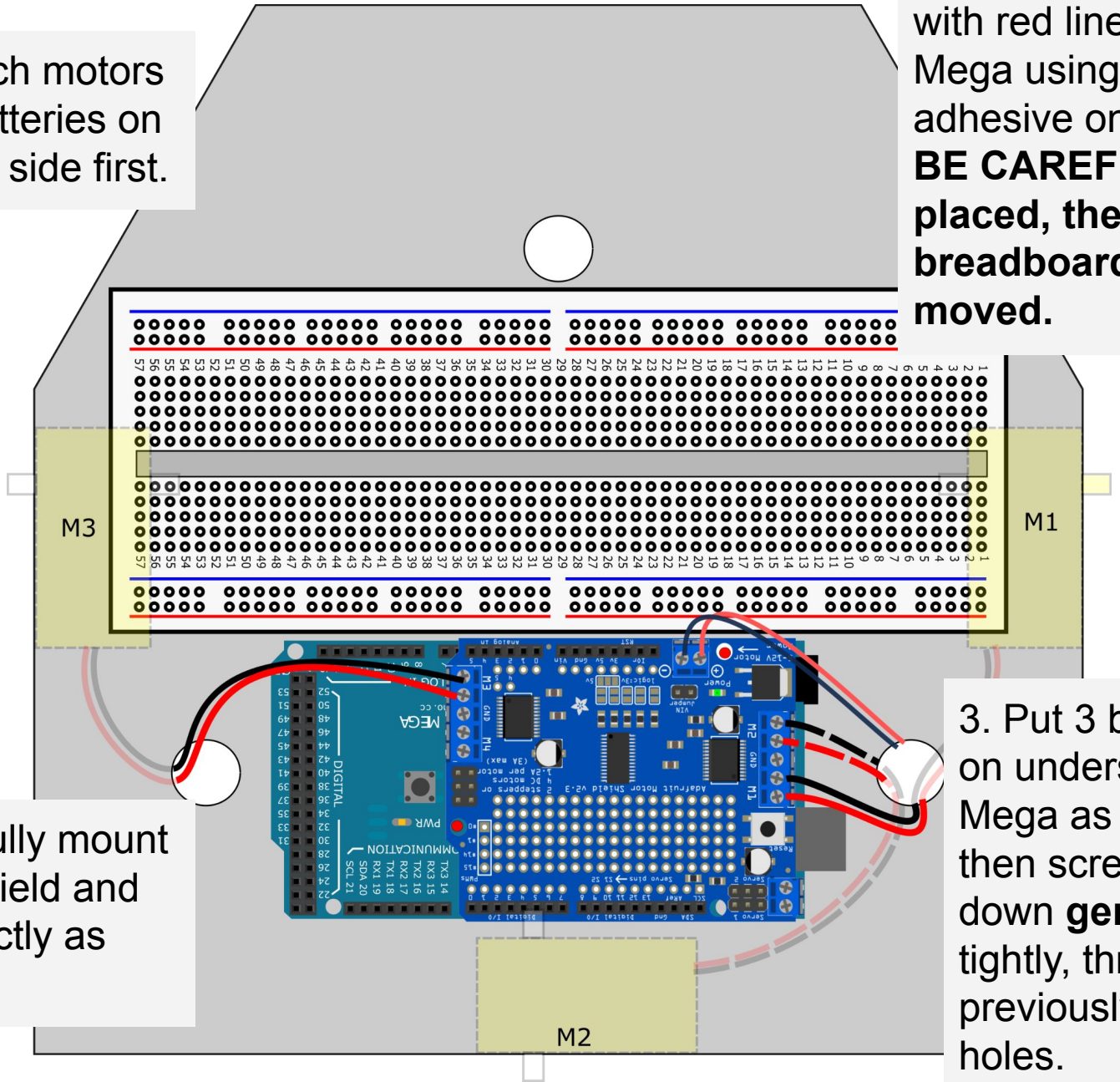
Omniwheel assembly (demo)

- Place motors into motor mounts. Make sure the stub on side of motor fits into hole on side of motor mounts.
- Lightly sand one side of center post of wheels and clean with alcohol.
- Lightly sand side of hubs with central round bump (leave the bump) and clean with alcohol.
- Using a SMALL amount of hot glue, glue the bump side of hub to the cleaned center post of wheel. The bump should help center the hub and wheel. Use screw to help with alignment, but don't’ attach to motor yet.

B3 Begin Assembly (shield demo)

1. Attach motors and batteries on bottom side first.

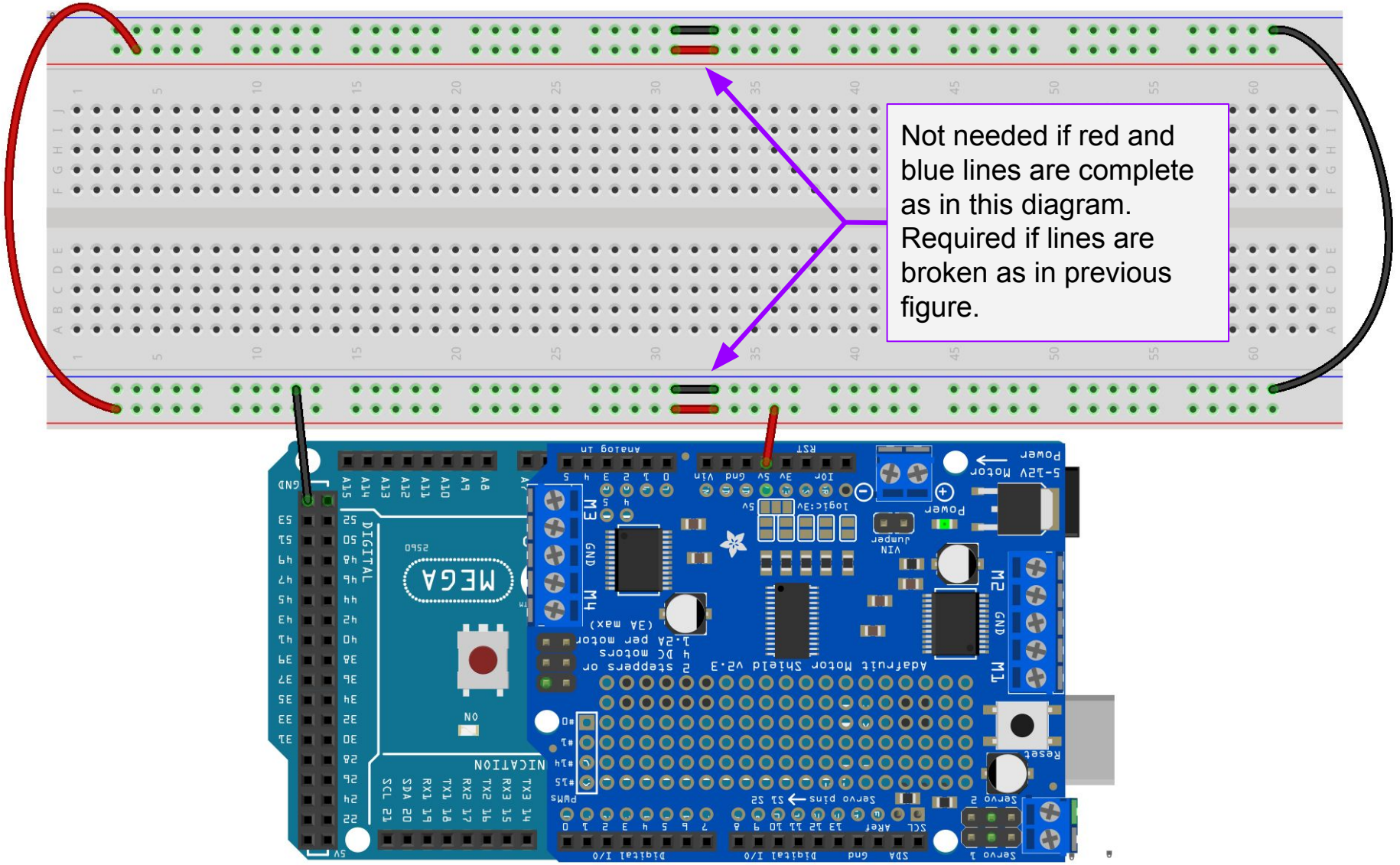
2. Attach breadboard with red line next to Mega using the adhesive on its back. **BE CAREFUL! Once placed, the breadboard can not be moved.**



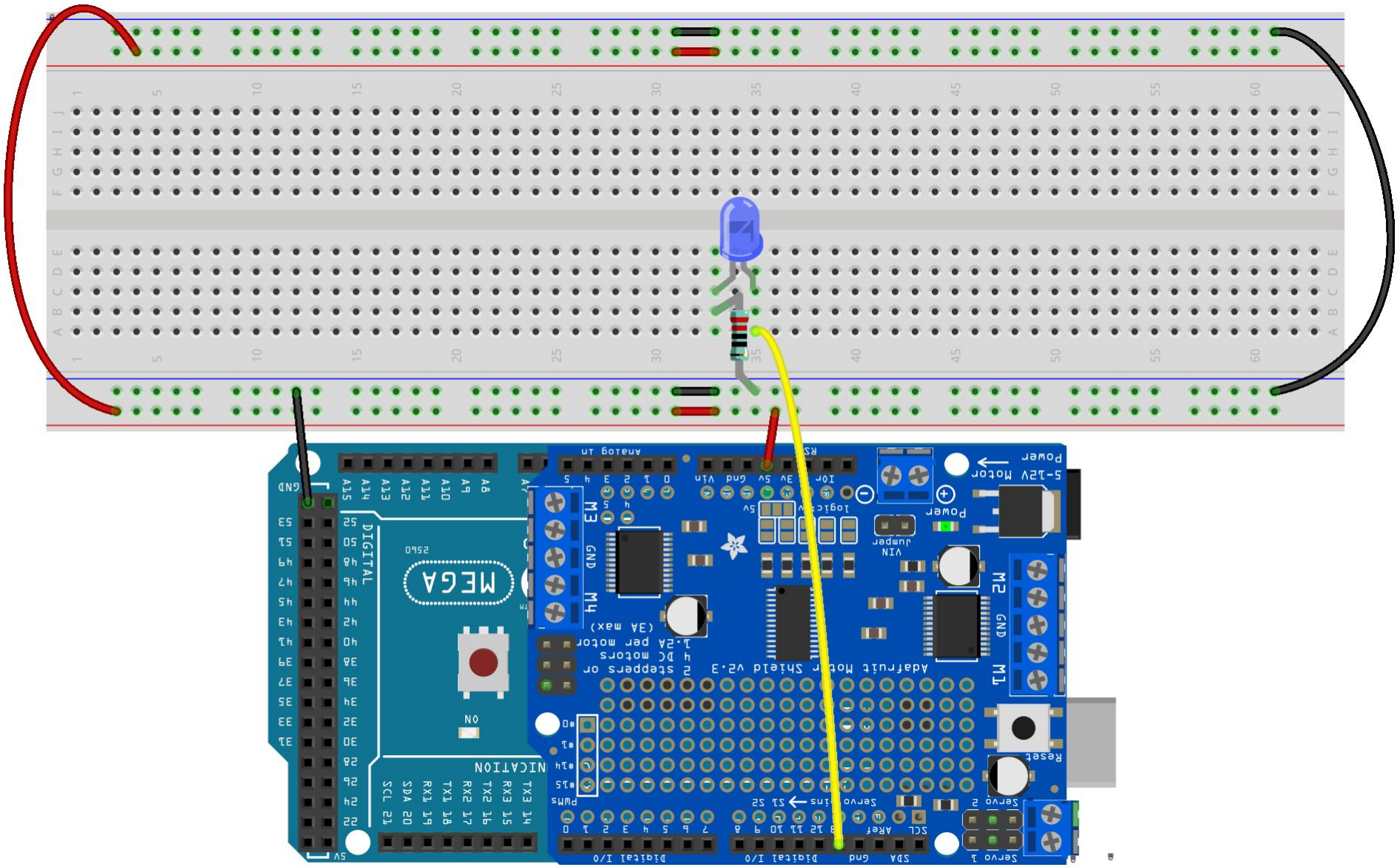
3. Put 3 balls of putty on underside of Mega as cushion, then screw Mega down **gently**, but not tightly, through previously drilled holes.

4. Carefully mount motor shield and wire exactly as shown.

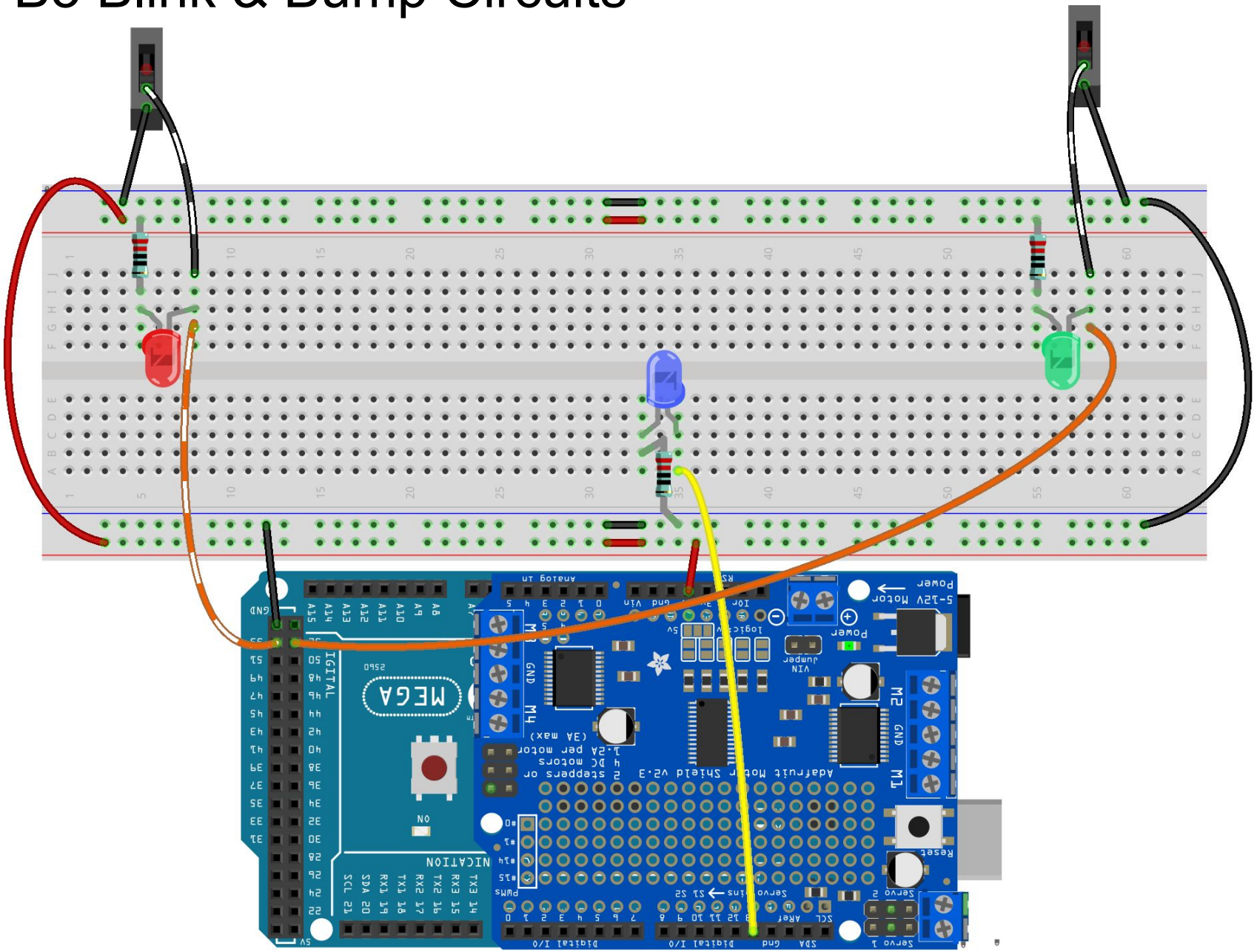
B3 Power and Ground Circuits



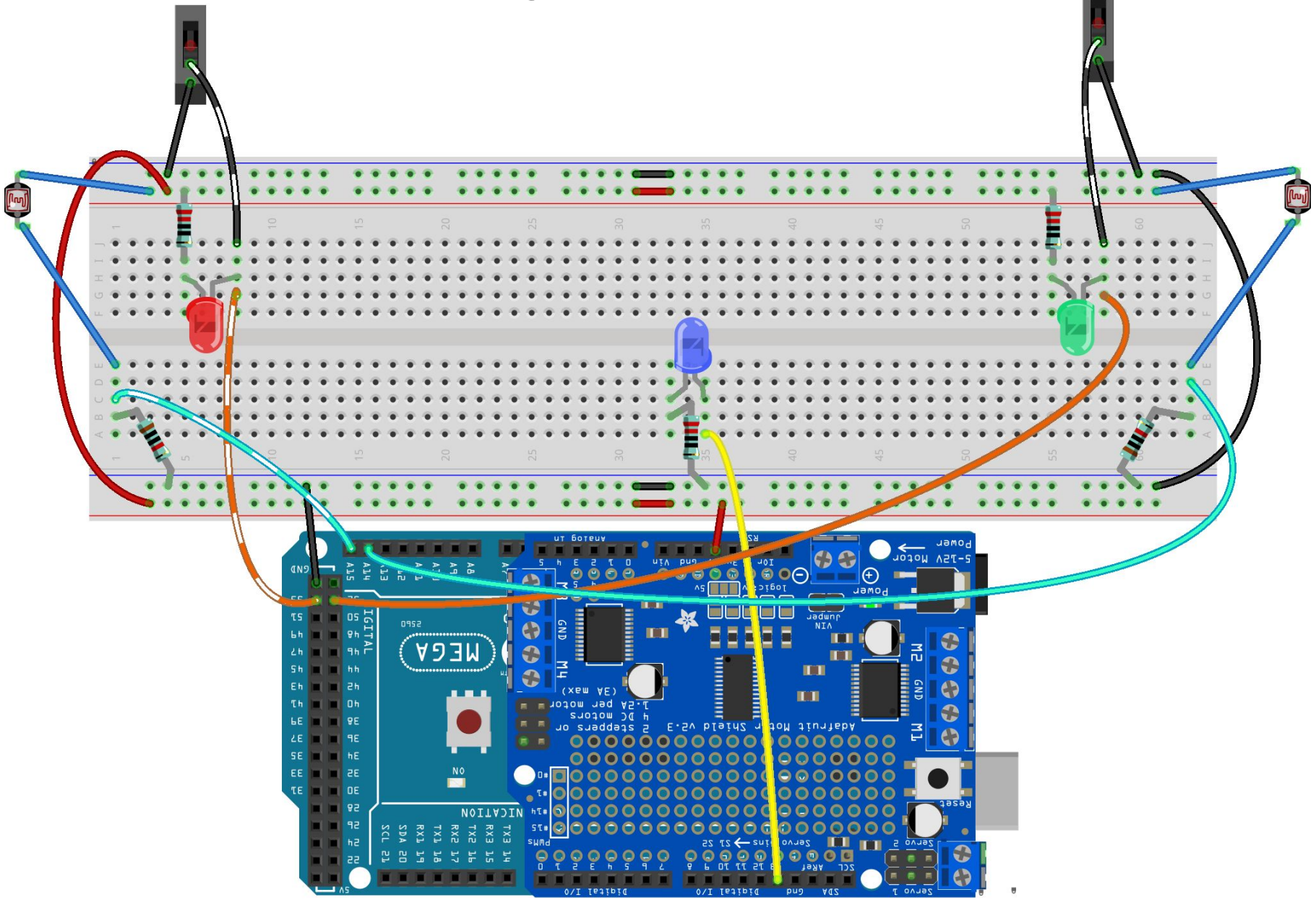
B3 Blink Circuits



B3 Blink & Bump Circuits

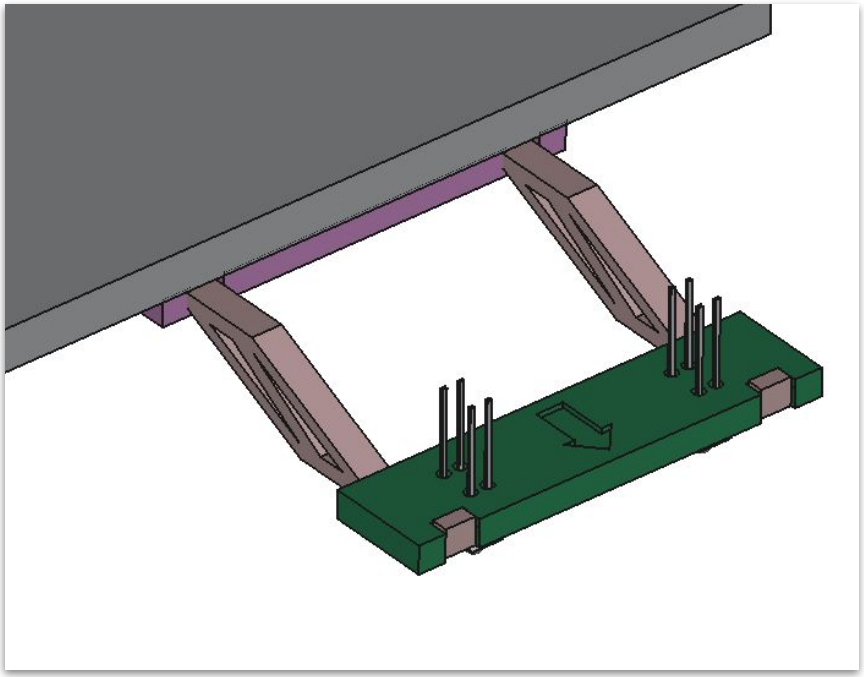


B3 Blink, Bump & Light Detection Circuits

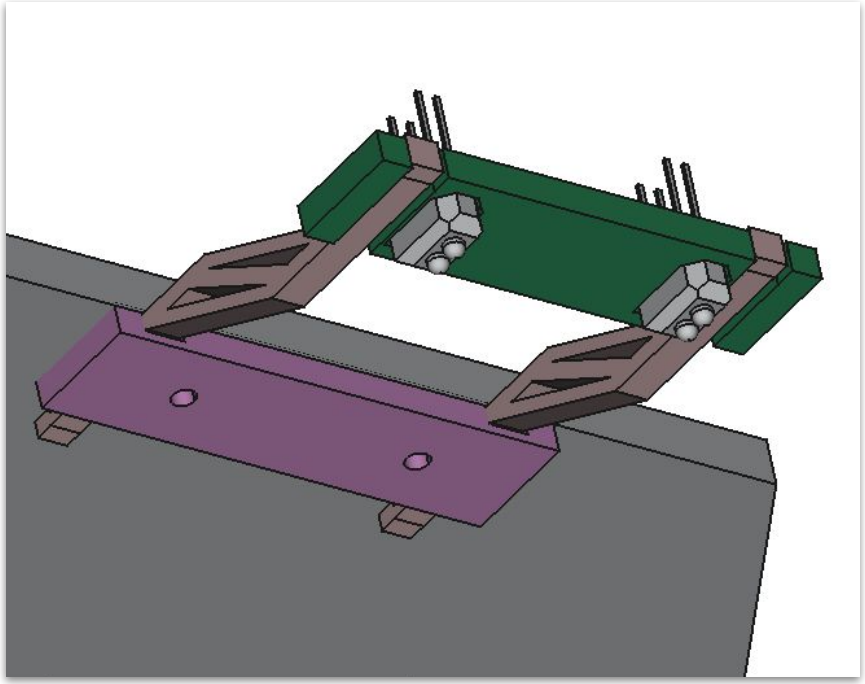


B3 Line follower assembly and orientation

Top View

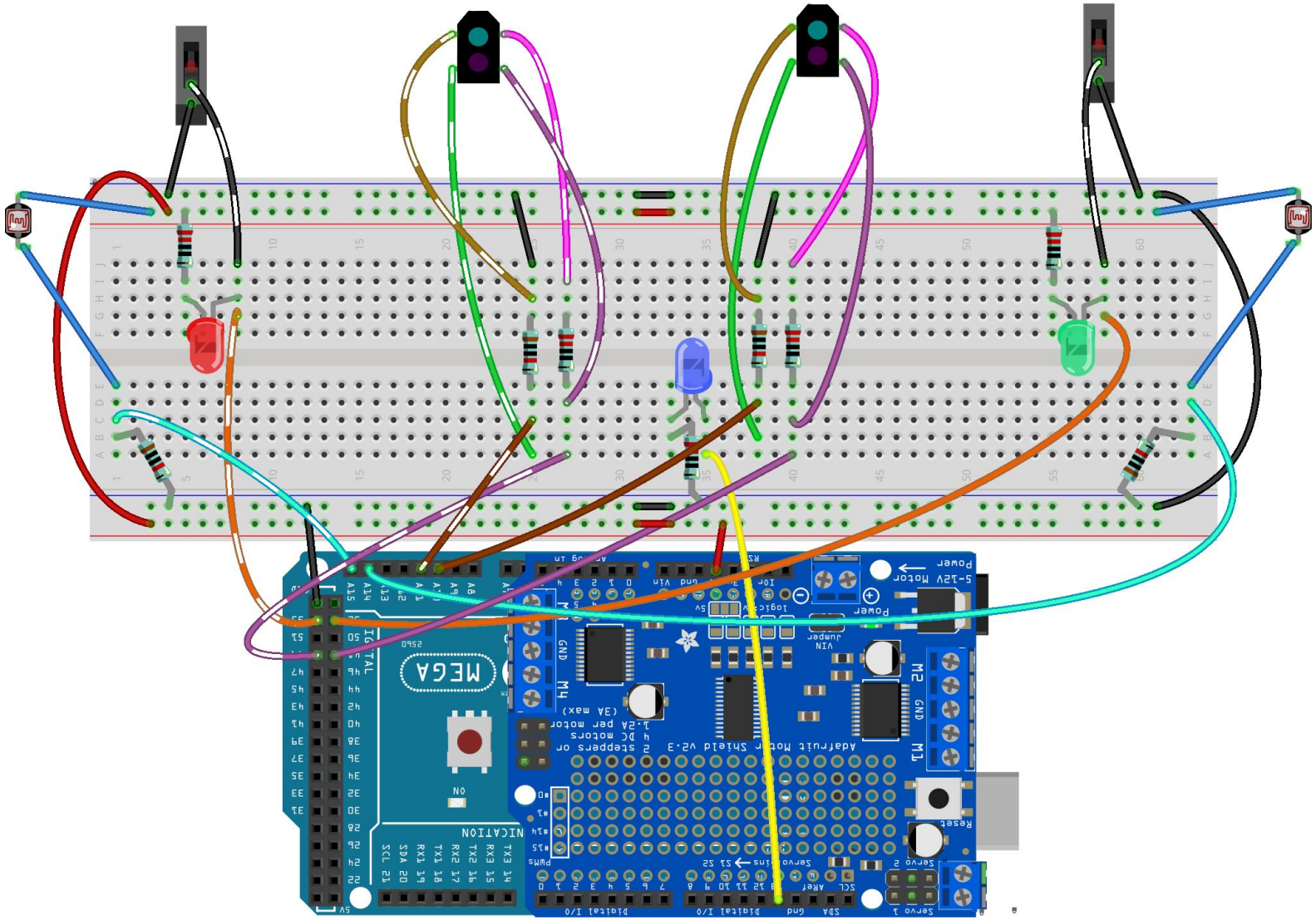


Bottom View



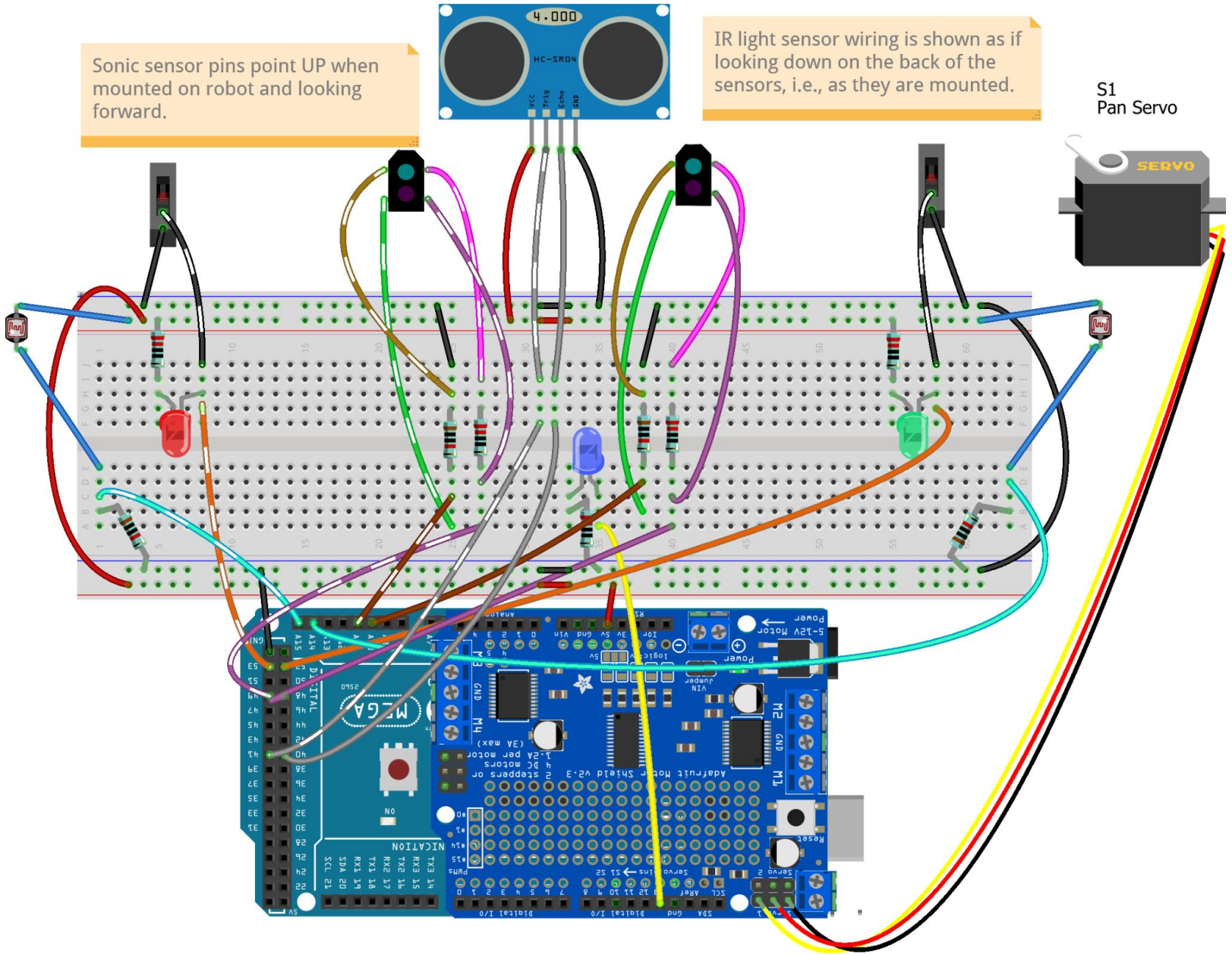
B3 Blink, Bump, Light & IR Circuits

IR light sensor wiring is shown as if looking down on the back of the sensors, i.e., as they are mounted.



fritzing

B3 Blink, Bump, Light, IR, Sonic & Servo Circuits



B3 Bonus Circuits

Sonic sensor pins point UP when mounted on robot and looking forward.

IR light sensor wiring is shown as if looking down on the back of the sensors, i.e., as they are mounted.

